



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF PHARMACY

Diploma Programme

(Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED “Diploma in Pharmacy”

Disclaimer: The diploma programme in Diploma in Pharmacy syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Pharmacy, Bhabha University

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

Visit us at: www.bhabhauniversity.edu.in, Mail Id info@bhabhauniversity.edu.in

Diploma in Pharmacy

II year

PLAN AND SCHEME OF EXAMINATION FOR THE DIPLOMA IN PHARMACY

(Based on effective teaching for 180 working days in one academic year)

Diploma in Pharmacy (Part-II) Examination

Subject Code	Subject	Max. Marks in Theory			Max. Marks in Practical		
		Examination	Sessional	Total	Examination	Sessional	Total
DP-201	Pharmaceutics-II	80	20	100	80	20	100
DP-202	Pharmaceutical Chemistry-II	80	20	100	80	20	100
DP-203	Pharmacology & Toxicology	80	20	100	80	20	100
DP-204	Pharmaceutical Jurisprudence	80	20	100	----	----	-----
DP-205	Drug Store & Business Management	80	20	100	----	----	-----
DP-206	Hospital & Clinical Pharmacy	80	20	100	80	20	100
Total				600			400

Scheme, Diploma in Pharmacy, Bhabha University, Bhopal

Pharmaceutics-II (201)

Paper Code 1949, Theory (75 Hours)

1. DISPENSING PHARMACY:

(i) Prescription:

Reading and understanding of prescriptions; Latin terms commonly used (Detailed study is not necessary), Modern methods of prescribing, adoption of metric- system Calculations involved in dispensing.

(ii) Incompatibilities in Prescriptions:

Study of various types of incompatibilities - Physical, Chemical and therapeutic.

(iii) Posology:

Dose and dosage of drugs, Factors influencing, Dose, Calculation of doses on the basis of age, sex and surface area, Veterinary doses.

2. DISPENSED MEDICATIONS:

Note: A detailed study of the following dispensed medication is necessary. Methods of preparation with theoretical and practical aspects, use of appropriate containers and closures. Special labelling requirement and storage conditions should be highlighted

(i) Powders:

Types of powder, Advantages and disadvantages of powders, Granules, Cachets and Tablet triturates. Preparation of different types of powders encountered in prescriptions. Weighing methods, possible errors in weighing, minimum weighable amounts and weighing of a material below the minimum weighable amount, Geometric dilution and proper usage and care of dispensing balance.

(ii) Liquid Oral Dosage Forms:

(a) Monophasic:

Theoretical aspects including commonly used vehicles, essential adjuncts like stabilizers, colourants and flavourants and flavours, with examples. Review of the following monophasic liquids with details of formulation and practical methods.

Liquids for internal liquids for external administration used on mucus membranes

Mixtures and Concentrates Gargles Syrups Mouth Washes Throat Paints Douches Elixirs Nasal Eye drops and Sprays Liniments Lotions

(b) Biphasic Liquid Dosage Forms:

(i) Suspensions (elementary Study): Suspensions containing diffusible solids and liquids and their preparation. Study of the adjuvant used like thickening agents, wetting agents, their necessity and quantity to be incorporated. Suspensions of precipitate forming liquids like tinctures, their preparation and stability. suspensions produced by chemical reaction. An introduction to flocculated/non-flocculated suspension system.

(ii) Emulsions: Types of emulsions, identification of emulsion system, formulation of emulsions, selection of emulsifying agents. Instabilities in emulsions. Preservation of emulsions.

(iii) Semi-Solid Dosage Forms:

(a) Ointments: Types of ointments, Classification and selection of dermatological vehicles. Preparation and stability of ointments by the following processes:

(i) Trituration

(ii) Fusion

(iii) Chemical reaction

(iv) Emulsification

(b) Pastes: Differences between ointments and pastes, Bases of pastes. Preparation of pastes and their preservation.

(c) Jellies: An introduction to the different types of jellies and their preparation.

(d) An elementary study of poultice.

(e) Suppositories and Pressaries - Their relative's merits and demerits, Types of suppositories, suppository bases, classification, properties, preparation and packing of suppositories use of suppositories for drug absorption.

(iv) Dental and Cosmetic Preparations: Introduction to Dentrifices, Facial cosmetics, Deodorants, Antiperspirants, Shampoos, Hair dressings and Hair removers.

(v) Sterile Dosage Forms:

(a) Parenteral Dosage Forms: Definition, General requirements for parenteral, dosage forms. Types of parenteral formulations, vehicles, adjuvants, processing, personnel, facilities and quality control. Preparation of Intravenous fluids and admixtures- Total parenteral nutrition, Dialysis fluids.

(b) Sterility testing, Particulate matter monitoring faulty seals packaging.

(c) Ophthalmic Products: Study of essential characteristics of different ophthalmic preparation. Formulation additives/special precautions in handling and storage of ophthalmic products.

PHARMACEUTICS-II

List of Practical (100 Hours)

Dispensing of at least 100 products covering a wide range of preparation such as mixtures, emulsions, lotions, liniments, E.N.T. preparation, ointments, suppositories, powders, incompatible prescription etc.

Books Recommended :(Latest Editions)

1. Indian Pharmacopoeia.
2. British Pharmacopoeia.
3. National Formulaires (N. F. I., B. N. F.)
4. Remington's Pharmaceutical Sciences.
5. Martindale's Extra Pharmacopoeia.

Pharmaceutical Chemistry-II (202)

Paper Code 1950, Theory (100 Hours)

1. Introduction to nomenclature of organic chemical system with particular reference to Hetero-Cyclic System containing up to 3 rings.

2. The chemistry of following Pharmaceutical organic compounds covering their nomenclature, chemical structure, uses and the important Physical and Chemical properties (Chemical structure of only those compounds marked with asterisk (*). The stability and storage conditions and the different types of Pharmaceutical formulations of these drugs and their popular brand names.

ANTISEPTICS AND DISINFECTANTS: Proflavine *, Benzalkonium chloride, Cetrimide, Chlore Cresol *, Chloroxylene, Formaldehy solution, Hexachlorophene, Liquified phenol, Nitro funantoin.

SULFONAMIDES:

Sulfadiazine *, Sulfaguanidine *, Phthaly sulfathiazole, Succinyl sulfathiazole, Sulfadimethoxine, Sulfamethoxy pryidazine, Sulfa methoxazole, Co-trimoxazole, Sulfacetamide

ANTILEPROTIC DRUGS:

Clofazimine, Thiambutosine, dapsone *, Solapsone. ANTI-TUBERCULAR DRUGS : Isoniazid *, PAS *, Streptomycin, Rifampicin, Ethambutol *, Thiacetazone, Ethionamide, Cycloserine, Pyrazinamide *.

ANTIAMOEBIAC AND ANTHELMINTIC DRUGS:

Emetine, Metronidazole *, Halogenated hydroxyquinolines, Diloxanide furoate, Paromomycin Piperazine *, Mebendazole, D.E.C.*.

ANTIBIOTICS:

Benzyl, Penicillin*, Phenoxy methyl, Penicillin*, Benzathine penicillin, Ampicillin*, Cloxacillin, Carbenicillin, Gentamycin, Neomycin, Erythromycin, Tetracycline, Cephalexin, Cephaloridine, Cephalothin, Griseofulvin, Chloramphenicol.

ANTIFUNGAL AGENTS:

Linecylenic acid, Tolnaflate, Nystatin, Amphoterecin, Hamycin.

ANTIMALARIAL DRUGS:

Chloroquine*, Amodiaquine, Primaquine, Proguanil, Pryimethamine*, Quinine, Trimethoprim.

TRANQUILIZERS:

Chlorpromazine*, Prochlorperazine, Trifluoperazine, Thiothixene, Haloperidol*, Triperidol, Oxypertine, Chlordiazepoxide, Diazepam*, Lorazepam, Meprobamate.

HYPNOTICS:

Phenobarbitone*, Butobarbitone, Cyclobarbitone, Nitrazepam, Glutethimide*, Methypylon, Paraldehyde, Triclofossodium,

GENERAL ANAESTHETICS:

Halothane*, Cyclopropane*, diethyl Ether*, Metho-Hexital Sodium, Thiopental Sodium, Trichloro Ethylene.

ANTIDEPRESSANT DRUGS:

Amitriptyline, Nortriptyline, Imipramine*, Phenelzine, Tranyl Cypromine.

ANALEPTICS:

Theophylline, Caffeine*, Coramine*, Dextroamphetamine.

ADRENERGIC DRUGS:

Adrenaline*, Noradrenaline, Isoprenaline*, Phenylephrine, Salbutamol, Terbutaline, Ephedrine*, Pseudoephedrine.

ADRENERGIC ANTAGONIST:

Tolazoline, Propranolol*, Practalol.

CHOLINERGIC DRUGS:

Neostigmine*, Pyridostigmine, Pralidoxime, Pilocarpine, Physostigmine*.

CHOLINERGIC ANTAGONISTS:

Atropine*, Hyoscine, Homatropine, Propantheline*, Benztropine, Tropicamide, Biperiden.

DIURETIC DRUGS:

Furosemide*, Chlorothiazide, Hydrochlorothiazide, Benzthiazide, Urea*, Mannitol*, Ethacrynic Acid.

CARDIOVASCULAR DRUGS:

Ethyl nitrite*, Glyceryl Trinitrate, Alpha Methyl Dopa, guanethidine, Clofibrate, Quinidine.

HYPOGLYCEMIC AGENTS:

Insuline, Chlorpropamide*, Tolbutamide, Glibenclamide, Phenformin*, Metformin.

COAGULANTS AND ANTI COAGULANTS:

Heparin, Thrombin, Menadione*, Bishydroxycoumarin, Warfarin Sodium.

LOCAL ANAESTHETICS:

Lignocaine*, Procaine*, Benzocaine.

HISTAMINE AND ANTI-HISTAMINIC AGENTS:

Histamine, Diphen Hydramine*, Promethazine, Cyproheptadine, Mepyramine*, Pheniramine, Chlorpheniramine*.

ANALGESICS AND ANTI-PYRETICS:

Morphine, Pethidine*, Codeine, Methadone, Aspirin*, Paracetamol*, Analgin, Dextropropoxyphene, Pentazocine.

NON-STERIODAL ANTI-INFLAMMATORY AGENTS:

Indomethacin*, Phenylbutazone*, Oxyphen Butazone, Ibuprofen.

THYROXINE AND ANTI-THYROIDS:

Thyroxine*, Methimazole, Methyl Thiouracil, Propylthiouracil.

DIAGNOSTIC AGENTS:

Iopanoic Acid, Propylidone, Sulfobromophtha Sodium, Indigotindisulfonate Sodium (Indigo Carmine), Evans Blue, Congo Red, Fluorescein Sodium*, Anticonvulsants, Cardiac Glycosides, Antiarrhythmic Antihypertensives and Vitamins.

STEROIDAL DRUGS:

Betamethazone, Cortisone, Hydrocortisone, Prednisolone, Progesterone, testosterone, Oestradiol Nandrolone.

ANTI-NEOPLASTIC DRUGS:

Actinomycins, Azathioprine, Busulphan, Chlorambucil, Cisplatin Cyclophosphamide, Daunorubicin, Hydrochloride, Fluorouracil, Mercaptopurine, Methotrexate, Mytomycin.

PHARMACEUTICAL CHEMISTRY-II

List of Practical (75 Hours)

1. Systematic qualitative testing of organic drugs involving Solubility determination, melting point and/or boiling point, detection of elements and functional groups (10 Compounds).
2. Official identification tests for certain groups of drugs included in the I.P. like Barbiturates, Sulfonamides, Phenothiazines, Antibiotics etc. (8 Compounds).
3. Preparation of three simple organic preparation.

BOOKS RECOMMENDED :(Latest Editions)

1. Pharmacopoeia of India.
2. British Pharmaceutical Codex.
3. Martindale's Extra Pharmacopoeia.

Pharmacology and Toxicology (203)

Paper Code 1951, Theory (75 Hours)

- 1.** Introduction to Pharmacology, scope of Pharmacology.
- 2.** Routes of administration of drugs, their advantages and disadvantages.
- 3.** Various processes of absorption of drugs and the factors affecting them Metabolism, distribution and excretion of drugs.
- 4.** General mechanism of drugs action and the factor which modify drug action.
- 5.** Pharmacological classification of drugs. The discussion of drugs should emphasise the following aspects:
 - (i)** Drugs acting on the Central Nervous System:
 - (a)** General anaesthetics, adjunction to anaesthesia, intravenous anaesthetics.
 - (b)** Analgesic, antipyretic and Non-Steroidal, anti-inflammatory drugs, Narcotic analgesics. Antirheumatic and antigout remedies. Sedatives and Hypnotics, Psychopharmacological agents, anti convulsants, analeptics.
 - (c)** Centrally action muscle relaxants and antiparkinsonism agents.
 - (ii)** Local anaesthetics.
 - (iii)** Drugs acting on autonomic nervous system.
 - (a)** Cholinergic drugs, Anticholinergic drugs, Anticholinesterase drugs.
 - (b)** Adrenergic drugs and Adrenergic receptor blockers.
 - (c)** Neurone blockers and Ganglion blockers.
 - (d)** Neuromuscular blockers, Drugs used in Myasthenia gravis.
 - (iv)** Drugs acting on eye, Mydriatics, Drugs used in glaucoma.
 - (v)** Drugs acting on respiratory system - Respiratory stimulants, Bronchodilators, Nasal decongestants, Expectorants and Antitussive agents.
 - (vi)** Antacids, Physiological role of histamine and serotonin, Histamine and Antihistamines, Prostaglandins.
 - (vii)** Cardio Vascular Drugs, Cardiotonics, Antiarrhythmic agents, Antianginal agents, Antihypertensive agents, Peripheral Vasodilators and drugs used in Atherosclerosis.
 - (viii)** Drugs acting on the blood and blood forming organs. Haematinics, Coagulants and Anticoagulants, Haemostatics, Blood substitutes and Plasma expanders.
 - (ix)** Drugs effecting renal function - Diuretics and Antidiuretics.
 - (x)** Hormones and Hormone Antagonists - Hypoglycemic Agents, Antithyroid drugs, Sex hormones and Oral contraceptives corticosteroids.
 - (xi)** Drugs Acting on Digestive System - Carminatives, Digestants Bitters, Antacids and drugs in pepticulcer, Purgatives and Laxatives, Antidiarrhoeals, Emetics, Antiemetics, Antispasmodics.

6. Chemotherapy of microbial diseases : Urinary antiseptics, Sulphonamides, Penicillins, Streptomycin, Tetracyclines and other antibiotics. Antitubercular agents, Antifungal agents, Antiviral drugs, Antileprotic drugs.
7. Chemotherapy of protozoal diseases. Anthelmintic drugs.
8. Chemotherapy of Cancer.
9. Disinfectants and Antiseptics. A detailed study of the action of drugs on each organ is not necessary.

List of Practical (50 Hours)

The first Six of the following experiments will be done by the students while the remaining will be demonstrated by the Teacher

1. Effect of K⁺, Ca⁺, Acetyl Choline and Adrenaline on Frog's heart.
2. Effect of Acetyl Choline on rectus abdominals muscle of Frog and Guinea Pig Ileum.
3. Effect of Spasmogens and relaxants on Rabbits intestine.
4. Effect of local anaesthetics on Rabbits Cornea.
5. Effect of mydriatics and mlotics on Rabbits Eye.
6. To study the action of Strychnine on Frog.
7. Effect of digitalis on Frog's heart.
8. Effect of hypnotics in mice.
9. Effect of convulsants and anticonvulsant in mice or rats.
10. Test for Pyrogens.
11. Taming and Hypnosis potentiating effect chlorpromazine in Mice/Rats.
12. Effect of diphenhydramine in experamentally produced asthma in guinea pigs.

Hospital and Clinical Pharmacy (204)

Paper Code 1954, Theory (75 Hours)

PART-I HOSPITAL PHARMACY:

1. Hospital - Definition, Function, Classifications based on various criteria, organization, Management and health delivery, System in India.
2. Hospital Pharmacy:
 - (a) Definition
 - (b) Functions and objective of Hospital, Pharmaceutical services.
 - (c) Location, Layout, Flow chart of materials and men.
 - (d) Personnel and facilities requirements including equipments based on individual and basic needs.
 - (e) Requirements and abilities required for Hospital pharmacists.
3. Drug Distribution system in Hospitals:
 - (a) Out patient services.
 - (b) In patient services -
 - (i) Types of services
 - (ii) Detailed discussion of Unit Dose System, Floor Ward Stock System, Satellite Pharmacy Service, Central Sterile Services, Bed Side Pharmacy.
4. Manufacturing:
 - (a) Economical considerations, estimation of demand.
 - (b) Sterile manufacture - Large and small volume parenteral facilities requirements, Layout, Production planning, man-power requirements.
 - (c) Non-Sterile Manufacture - Liquid orals, Externals, Bulk concentrates.
 - (d) Procurement of stores and testing of raw materials.
5. Nomenclature and uses of surgical instruments and hospital equipments and health accessories.
6. P. T. C. (Pharmacy Therapeutic Committee), Hospital Formulary System and their organization, Functioning, Composition.
7. Drug Information service and Drug Information Bulletin.
8. Surgical dressing like cotton, gauze, bandages and adhesive tapes including their pharmacopoeial tests for quality. Other hospital supply e.g. I.V. sets, B. G. sets, Ryals Tubes, Catheters, Syringes etc.
9. Application of computers in maintenance of records, inventory, medication monitoring, drug information and data storage and retrieval in hospital and retail pharmacy establishments.

PART-II CLINICAL PHARMACY:

1. Introduction to Clinical Pharmacy Practice - Definition, Scope.

2. Modern Dispensing Aspects - Pharmacists and Patient counseling and advice for the use of common drugs, medication history.
3. Daily terminology used in the practice of Medicine.
4. Disease manifestations and pathophysiology including salient symptoms to understand the disease like Tuberculosis, Hepatitis, Rheumatoid Arthritis, Cardio-Vascular diseases, Epilepsy, Diabetes, Peptic Ulcer, Hypertension.
5. Physiological parameters with their significance.
6. Drug Interactions:
 - (a) Definition and Introduction.
 - (b) Mechanism of Drug Interaction.
 - (c) Drug - Drug interaction with reference to Analgesics, Diuretics, Cardio Vascular Drugs, Gastro - Intestinal agent vitamins and hypoglycemic agents.
 - (d) Drug -Food interaction.
7. Adverse Drugs Reactions:
 - (a) Definition and significance.
 - (b) Drugs - Induced diseases and Teratogenicity.
8. Drugs in Clinical Toxicity - Introduction, General treatment of Poisoning, Systematic antidotes. Treatment of insecticide poisoning, Heavy metal poison, Narcotic drugs, Barbiturate, Organophosphorus poisons.
9. Drugs dependences, Drugs abuse, Addictive drugs and their treatment, Complications.
10. Bio-availability of drugs, including factors affecting it.

HOSPITAL AND CLINICAL PHARMACY

List of Practical (50 Hours)

1. Preparation of transfusion fluids.
2. Testing of raw materials used in (1).
3. Evaluation of surgical dressings.
4. Sterilization of surgical instruments, glass ware and other hospital supplies.
5. Handling and use of data processing equipments.

BOOKS RECOMMENDED (Latest Editions)

1. Romington's Pharmaceutical Sciences.
2. Martindale's Extra Pharmacopoeia.

Drug Store and Business Management (205)

Paper Code 1953, Theory (75 Hours)

PART - I COMMERCE (50 Hrs)

1. INTROCUCTION: Trade, Industry and Commerce, Functions and subdivision of Commerce, Introduction to Elements of Economic and Management.
2. Forms of Business organizations.
3. Channels of Distribution.
4. DRUG HOUSE MANAGEMENT: Selection of site, Space Lay-out and legal requirements. Importance and objectives of purchasing selection of suppliers, Credit information, Tenders, Contracts and Price determination and legal requirements thereto. Codification, Handling of drugs stores and other hospital supplies.
5. INVENTROY CONTRL: Objective and importance, Modern techniques like ABC, VED analysis, the lead time, Inventory carrying cost, Safety stock, Minimum and maximum stck levels, Economic order quantity, Scrap and surplus disposed.
6. Sales promotion, Market Research, Salesmanship, Quality of a salesman, Advertising and Window Display.
7. Recruitment, Training, Evaluation and Compensation of pharmacist.
8. BANKING AND FINANCE: Service and functions of bank, Finance Planning and sources of finance.

PART -II ACCOUNTANCY:(25 Hrs)

1. Introduction to the accounting concepts and conventions, Double entry, Book keeping, Different kinds of accounts.
2. Cash Book.
3. General Ledger and Trail Balance Sheet.
4. Profit and Loss Account and Balance Sheet.
5. Simple techniques of analysing financial statements.
6. Introduction to Budgeting.

BOOKS RECOMMENDED (Latest Editions)

1. Remington Pharmaceutical Sciences.

Pharmaceutical Jurisprudence (206)

Paper Code 1952, Theory (50 Hours)

1. Origin and nature of Pharmaceutical legislation in India, its scope and objectives. Evolution of "Concept of Pharmacy" as an integral part of the Health Care System.
2. Principles and significance of Professional Ethics. Critical study of the code of Pharmaceutical Ethics Drafted by Pharmacy Council of India.
3. Pharmacy Act, 1948 - The General study of the Pharmacy Act with special reference to Education Regulations, working of State and Central Councils, constitution of these councils and functions. Registration procedures under the Act.
4. The Drugs and Cosmetics Act 1940 - General study of the Drugs and Cosmetics Act and the Rules thereunder. Definitions and salient features related to retail and whole sale distribution of drugs. The powers of Inspectors, the sampling procedures and the procedures and formalities iontaining licenses under the rule. Facilities to be provided for running a Pharmacy effectively. General study of the schedules with special reference to schedules C,C1,F,G,J,H,P and X and salient features of lanelling and storage conditions of drugs.
5. The Drugs and Magic Remedies (Objectionable Advertisement) Act, 1954 - General study of the Act, objectives special reference to be laid on Advertisements, Magic remedies and objectionable and permitted advertisements - diseases which can not be claimed to be cured.
6. Narcotic Drugs and Psychotropic Substances Act, 1985 - A brief study of the act with special reference to its objectives, offences and punishment.
7. Brief introduction to the study of the following acts :
 - (i) Latest Drugs (Price Control) Order in force.
 - (ii) Poisons Act, 1919 (as amended to date).
 - (iii) Medicinal and Toilet Preparations (Excise Duties) Act, 1955 (as amended to date).
 - (iv) Medical Termination of Pregnancy Act, 1971 (as amended to date).

BOOKS RECOMMENDED (Latest Editions)

Bore Acts of the said published by the Government.